Flight Dynamics of the X-57 Maxwell in JSBSim

A built-up model with fourteen propellers, proved on the Tecnam P2006T and checked against published CFD, tests and simulation

Technical report • November 2026

Abstract

A flight-dynamics model of the X-57 Maxwell in its Mod III and Mod IV forms has been written in JSBSim for the FlightGear simulator. Mods III and IV never flew, so nothing here can be compared with a flight of those aircraft: the references are the CFD database, wind tunnel and vibration tests of their parts, and piloted-simulation results, all published. The model is built up part by part, 17 wing strips a side, 3 stabilator strips a side, 4 fin strips and 11 fuselage stations, so that any one of the fourteen propellers can fail and leave its own part of the wing. Each wing strip sits in the slipstream of its own disc, carried by the ring-vortex wake used in the same author's helicopter models with one fitted constant, in place of the three a plain jet needed. The method was first proved on the Tecnam P2006T, whose fuselage and tail the X-57 keeps: it passes that aircraft's flight manual, cruise within 2.7 kt at 27 points and climb on one and two engines within limits. Its flight-test misses are traced to the test paper's methods by a second source: at the Mod II simulator's mass the model gives that simulator's short period within 5 %, and the paper's own derivatives give 4.95 rad/s where its flight figure is 3.41. On the X-57 the model reproduces the CFD's clean and flapped lift, the propellers' added lift inside the solvers' own spread, and cruise L/D of 13.6 against 13.5. Its modes follow the published simulator's within 0.03 of damping and 10 % of frequency for Mod III. A motor lost with no pilot follows the published time histories: at 3 s bank $-79°$, pitch $-10.8°$, heading $-23.2°$ against about $-87°$, $-11°$, $-23°$, once the run is flown in the configuration the published traces show rather than the one their text states. All limits were set before the results and none was moved: 199 checks pass and 41 miss. Every miss is listed with its cause or marked open. The largest are climb after a motor failure with distributed thrust, 69 against 360 ft/min, and three frequencies against the Mod II simulator.

1 Introduction

The X-57 Maxwell was a battery-electric demonstrator built on a Tecnam P2006T [1]. Mod II replaced the two piston engines with electric motors in the same places. Mod III replaced the wing with one of 42 % of the area, aspect ratio 15, and moved the cruise motors to the wingtips, where their propellers turn against the tip vortices. Mod IV powers twelve folding high-lift propellers along the leading edge, which Mod III carries folded and inactive [1], to bring the stall speed back to the P2006T's [2]. The project ended with Mod II built and tested on the ground, the Mod III wing built and vibration tested [3], and neither of the later forms flown.

That leaves an unusual problem for a flight model: a large and open body of data, with no flight to hold it to. The CFD database [1] covers the wing, flaps and propellers; the high-lift propeller was tested in a wind tunnel [4]; the batteries, motors, controllers and cockpit are documented [6, 7, 8]; and the piloted simulator's results are published [10, 11, 12, 13]. All of those except the tests are predictions. Holding one prediction to another shows only agreement, so each check here is labelled by what it is held to: **M** measured, **C** CFD, **S** a published simulator, **D** a design document.

A table model on the database's own structure reproduces the CFD by construction, and is kept as the reference. It can't lose one motor's slipstream from one part of the wing except through the database's increments. The model that ships is built up instead: strips, discs, tail and bodies, whose failures come out of the parts.

2 Reference data

When sources disagree the shipped model follows the highest-ranked: flight test of that aircraft with a clean method, then wind tunnel at a comparable Reynolds number and configuration, then CFD, then a published simulator, then handbook estimates. A simulator ranks low because it's built from the others. The model is never tuned towards two sources at once. A comparison with a simulator is run on that simulator's inputs (its mass, its trim, its configuration), so that what's left is the model's own error; a part that vanishes was only a difference in data. Rank doesn't excuse a source from a sanity check: section 5.1 sets aside a higher-ranked figure for a stated reason.

3 The model

3.1 Implementation

One generator writes the Mod III, Mod IV and P2006T builds as plain JSBSim XML, with no scripting in the flight model. Forces and moments go through the aerodynamics axes. The build runs at 120 Hz, a rate at which the bending modes, flights and failures have converged. Inside FlightGear the force graph has to run as a system channel with stored outputs: read through FlightGear's property tree it would be worked out again at each read, and the start never finishes. Every property read in FlightGear takes a lock, so the shipped graph is laid out to read few: functions read once are written into their reader, unread ones are dropped, and the high-lift channels rest while those propellers are stopped and disarmed. That takes Mod IV from about 6900 reads a step to 6600, or 5600 at rest. A check flies this build beside the plain one, and the two match to the last bit.

3.2 Airframe

The wing is 17 strips a side (fig. 2): 10 flapped, 4 with the aileron and 3 plain. Strips end at each propeller's edge, so that a disc's slipstream has strips of its own. Their sections from NeuralFoil on the geometry's own airfoils and a lifting-line solution for the spanwise loading. The slotted flap's section



Figure 1: The model in FlightGear: Mod IV in a banked turn with the high-lift propellers running.

Table 1: Sources and what they contribute.

Source	Content used
TM-20250001715 [1]	CFD database, power off and on, ground planes, contactor failures, model structure
NASA OpenVSP model	geometry of wing, nacelles, blades
20210016834, 20220006506 [4, 5]	high-lift propeller in the tunnel, blade weights
20240006496, 20230004174 [10, 11]	simulators: mass, modes, failures, Mod II's identification
20240006682, 20240010539 [12, 13]	motor failure histories, DiTTo climbs
20240011864, 20240007439 [3, 14]	Mod III wing vibration test, flutter model
ICD-CEPT-006 [6]	cockpit, switches, sequences, displays
20240009560, 20240005892, 20170005797 [8, 7, 9]	motors, controllers, power system
Chin et al. [16]	cell model, thermal
P2006T flight manual [18]	performance, stall, gear system
Nicolosi et al. [19]	P2006T flight tests, tunnel, derivatives

lift, moment and drag are fitted to the CFD at flaps 0, 10 and 30, power off, and flaps 20 is left out of the fit as a test. Tail, fin and bodies follow the method proved on the P2006T. Past 8° a pitching moment on body incidence carries what the strips can't, because the CFD's pitch break comes at the same incidence with the propellers on or off, where a blown section is still short of stall. Rate derivatives are carried from the strips' to the tunnel-based ones the simulator uses, because without that the roll modes run 19 to 34 % quick.

3.3 Propellers and motors

Propellers are discs, not blades: the blades of twelve propellers pass at 400 Hz, far beyond the frame. Each of the fourteen discs keeps its own speed. Thrust, torque and in-plane force come from offline blade-element tables on the published blades. The high-lift blades run about 10° folded forward on their hinges, so the tables use an effective 20.0° against 27.35° drawn: partly documented, partly fitted to the published schedule. The cruise propeller's speed is governed by its blade pitch, which an electric hub changes at 1°/s, and each motor is a torque balance on the rotor's inertia. No rotor inertia is published. The cruise rotor's 0.65 kg m^2 is about 0.5 for the propeller, from the manufacturer's 1997 data sheet (0.6 kg m^2 with 160 cm blades, scaled to 152 cm), and 0.15 for the motor's rotor, estimated; the range is 0.58 to 0.73. The high-lift rotor's 0.0085 comes from the measured blade weights and the reference motor's parts [15].

The cruise propellers turn against the tip vortices, inboard blade rising: the published statements [23, 21] and the published 3-D geometry, measured blade by blade, agree. Each wing's six high-lift propellers turn the same way as its cruise propeller, the wings mirrored. The published geometry builds them so. The Mod IV CFD's skin friction [21] has the same lopsided pattern in every bay of a wing and its mirror image on the other, where alternating propellers would flip it from bay to bay. And on the X-57's predecessor wing, CFD gave propellers all turning against the tip vortex 0.31 more lift coefficient than pairs turning towards each other [22]. The simulator paper's "each pair of motors was counter-rotating" [10] is taken as left against right. It can also be read as neighbours, as this model first did, so that simulator may differ.

3.4 Slipstream

The axial speed at each strip comes from a tube of ring vortices behind the disc, after Heyson and Katzoff [20], the model used for the same author's V-22, UH-60 and Bo 105. Its contraction, 0.90 of the radius, is momentum theory's. The slipstream's effect on a wing is then reduced by one fitted constant, on the jet's strength:

$$a_{\text{eff}} = a\left(1 - k\,\frac{a}{u+a}\right), \qquad k = 0.70 \tag{1}$$

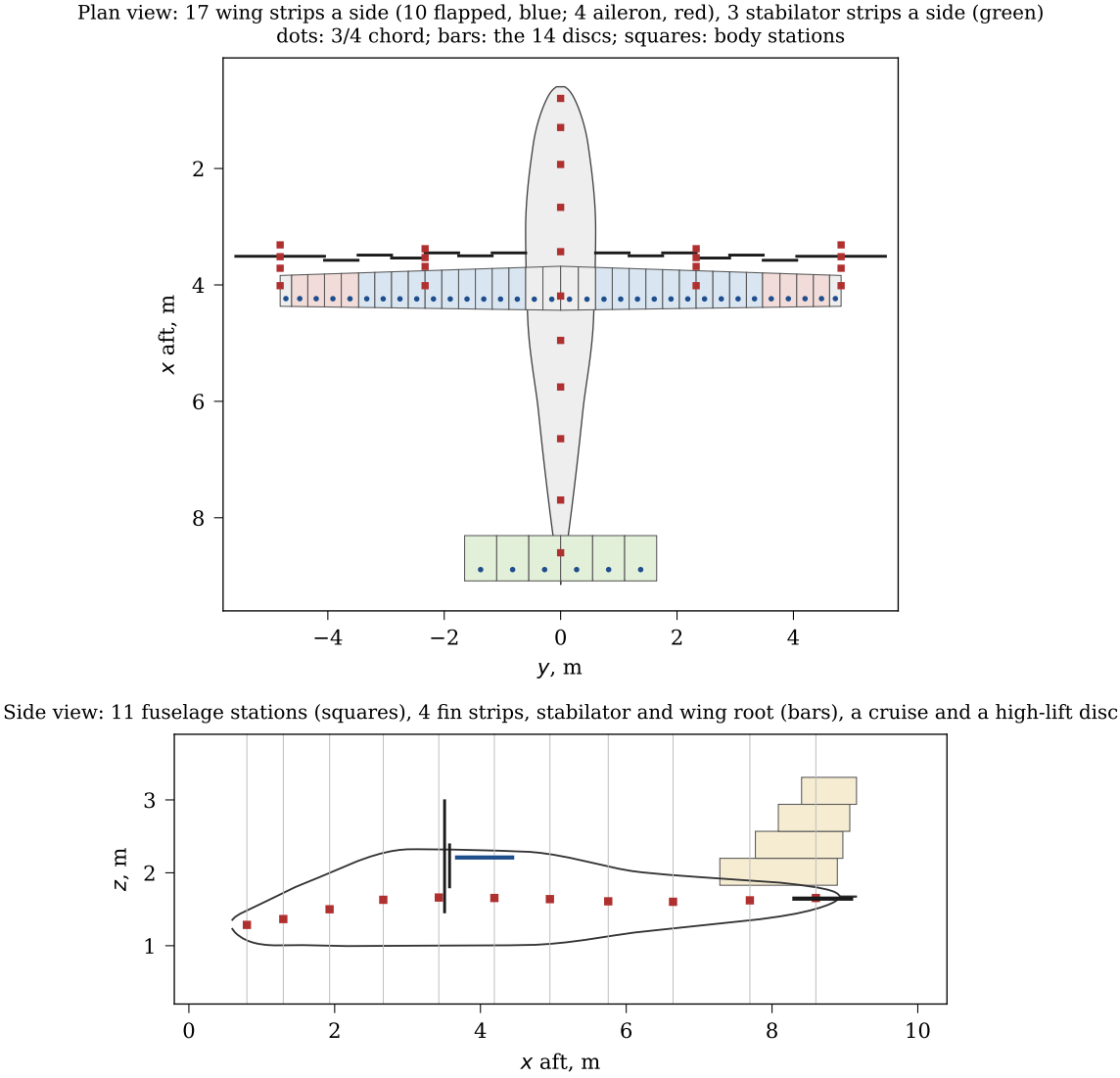


Figure 2: The model's discretisation, drawn from the generator's data, Mod IV. Strips end at each disc's edge, and at the flap's and aileron's. The P2006T's build has 13 wing strips a side (6 flapped, 3 aileron, 2 on the winglet), with the same 3 stabilator and 4 fin strips and 11 fuselage stations. The tip nacelles have 4 stations each; the high-lift nacelles are lumped, six to a side at their mean span, also on 4 stations.

It probably stands for a jet no taller than the chord, which does less than strip theory says; that isn't proved. The plain momentum jet this replaced needed three fitted constants for much the same result (section 5.5). No dynamic inflow is modelled: a Peters–He wake on each disc would give its own inflow, which the tables already hold in steady flow, and wouldn't change how the slipstream reaches the wing.

3.5 Named fits

Fits are named, and a fitted figure can't pass a check by itself. Besides the sections and those above: the flap's turning of the jet; the slipstream at the tail; a jet's lift from nearer the centre after a contactor failure; sideslip's roll and yaw on sideslip, incidence and flap, from the CFD, power off, because the strips lose the windward lift near stall where the CFD keeps it; one propeller's roll from the published tables, less the strips' own share, because a windmilling propeller takes lift off its own wing; the rudder, whose section table ran out towards full travel; ground effect on height, incidence and flap, power off and on; the gear's drag area of 4 ft^2, from the simulator's phugoid.

3.6 Power, battery and cockpit

The traction battery follows a cell model [16]. Seven avionics buses, the essential battery, two converters and their cross buses run inside the flight model, so that a shed bus or a failed converter reaches the contactors and the motors as the handbook says [6]. Voltages are fixed and loads estimated. The cockpit's logic is in the aircraft's own files; the companion manual describes it.

Table 2: All checks, by what they're held to: M measured, C CFD, S a simulator, D a design document.

Checks	Against	Pass	Miss
Modes and CFD increments, built-up model (limits written for the table model)	C, S	3	9
Propellers, motors, governor, battery	M, S	24	0
P2006T: flight manual, flight tests, tunnel, Mod II simulator	M, S	47	17
X-57 built-up aerodynamics against CFD	C	30	2
Ground effect, climbs, failures, gear	C, S, M	38	11
Wing bending	M, S	30	2
Cockpit power and propeller modes	D	27	0
All		199	41

3.7 Wing bending and gear

The wing bends in its first mode, symmetric and antisymmetric, as in the same author's ASK 21: stiffness fitted to the Mod III wing's vibration test [3], with tip masses and without. The panels' coupling to the airframe is solved in the system, with an effective mass for each kind of motion. Tyre deformation drag follows the published equation [17], held at its value past 55 KIAS, where its fit ends. Gear terms reach the panels as the mean of two frames, which removes a 60 Hz shake between JSBSim's tyre friction and the panels' reaction.

3.8 Mass

FlightGear flies the published gross weights: 3000 lb for Mod III [1] and a projected 3200 lb for Mod IV [14]. Comparisons with the simulator run at its 3075 lb. The 28.6 % CG and the inertias are the simulator's [10]: no other is published. With no fuel, weight and CG stay constant through a flight.

4 Method

The work went in phases, each with limits written before its first result, taken from the sources' own spread where they have one. A limit isn't moved afterwards, and a miss stays printed until its cause is fixed. One script refits the model and runs every check, at four processes. Where a test's configuration turned out to be wrong it was changed and the old run kept printing beside it, with the reason given. Table 2 gives the count by group.

5 Results

5.1 The method on the P2006T

(M) The built-up method passes the P2006T's flight manual: 27 cruise points mean +1.3 kt, rms 1.7, worst 2.7 (fig. 3, one drag fit); climb within limits on two engines and one, 1129 against 1102 ft/min at maximum weight and 217 against 206 on one engine. Stall speeds are fitted. Against the tunnel, part by part, it gives lift slope 0.0825/deg (0.080) and the neutral point 41.4 % (41). In flight, phugoid 28.8 s (27), damping 0.088 (0.09); Dutch roll 2.96 s (3.25), damping 0.305 (0.26).

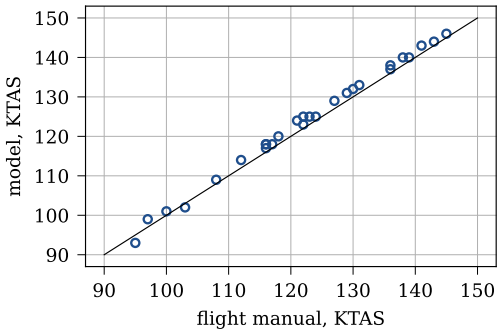


Figure 3: P2006T cruise speed at the manual's power, 27 points **(M)**.

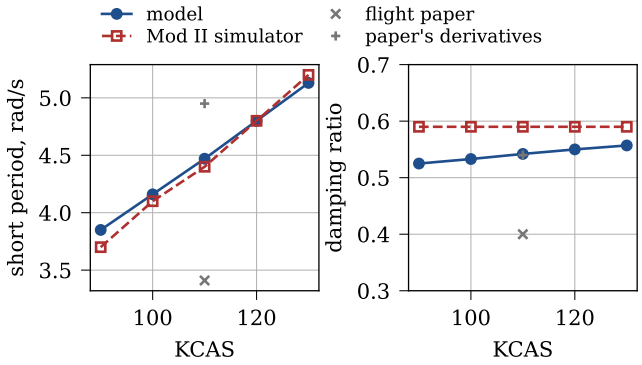


Figure 4: Short period at the Mod II simulator's mass against that simulator, read by eye **(S)**. The flight paper's measured figure and the one its own derivatives give, at 110 KIAS and its lighter mass and forward CG, are shown for scale.

Six checks against the flight paper [19] miss, and a second source says why. The simulator for Mod II was built on an identification of an instrumented P2006T and shown matching that aircraft's flight data [11], and its paper prints mass, CG and inertias, so the model can be run at them **(S)**. Its short period then agrees across the range (fig. 4): 3.85, 4.47, 5.13 rad/s against 3.7, 4.4, 5.2, damping 0.53 to 0.56 against 0.59. The flight paper's 3.41 rad/s and 0.40 are the outliers. Its maximum-slope method needs a true step, and the pilot's took about 0.6 s: the same method on the model's response to a 0.6 s step gives 3.68 rad/s, though the model's root is 5.92 at that mass and CG. The lift slope misses too, but the paper's 4.15/rad is the outlier again. Read together, because they share their points, the simulator paper's figures 11 and 12 give C_L against α: 5.09/rad, straight through all five cruise points. The nose vane appears to read about 1.3 of true incidence, and the simulator's authors replaced the published slope themselves [11]. The model's trimmed 5.70 is still 12 % above the simulator's, a real excess, and a miss. Aileron per degree of sideslip is 0.75 against the flight paper's 0.58 and, at the simulator's mass, 0.93 against its 0.80. The tunnel's $C_{Y\beta}$ of −0.69 stays open: sources run from −0.36 to −0.69, and the model gives −0.48, or −0.65 with winglets and power.

Against the same simulator three other frequencies miss. Dutch roll is 42 % high at 110 KCAS (1.80 against 1.27 rad/s), but scaling the flight test's 2.0 rad/s by the inertia gives 1.75, so measurement sides with the model. Phugoid is 17 to 24 % high, and the flight test's 27 s sides with the model again. The roll time constant is 0.1 s against the simulator's 0.7 to 0.9 s: any published C_{lp} at the printed inertia gives 0.1 to 0.2 s, and the flight paper calls the mode too short to observe. That gap

is unexplained.

None of this reaches the X-57's lift or stability, which come from the CFD: the P2006T is where the method was proved.

5.2 X-57 aerodynamics

(C) Power off, the fitted flaps match the solvers' mean within 0.003 in C_L and 0.020 in C_m; flaps 20, unfitted, within 0.055 and 0.026. Unfitted increments below stall: stabilator C_L 16 % and C_m 7 % rms, aileron roll 16 %, sideslip side force 8 %. Sideslip's moments and the rudder are fitted now and are no longer tests. Trimmed stall speeds for Mod III are 82.7, 80.6 and 70.8 KCAS against the published 88, 80, 73, worst 5.3, a narrow miss. At 150 KTAS L/D is 13.6 (13.5); the peak 14.9 (15.9) at 110 KCAS.

5.3 Propellers and battery

(M) The high-lift disc's tables are inside the tunnel's scatter: C_T rms 0.0064, C_Q 0.0007. The windmilling disc turns 2640 rpm at 75 KEAS, inside the measured 2483 and 2654. **(S)** Torque is within 7 % of the published schedule to $J = 1.1$, the cruise tables pass 37 published points to 0.0003, and the rpm schedule is exact to 0.2 rpm. The published slow-flight mission draws 41.9 kWh (41.9), lowest 378 V (370), depth of discharge 80.0 % (82.6), hottest pack 59.3 °C (57.9). The cell is within 0.09 V of the published figure.

5.4 Flight modes

(S) Against the simulator's linear models [10], Mod III is within 0.013 in phugoid damping, 0.032 short period, 0.012 Dutch roll, 5 % roll time constant, and 10 % frequency (fig. 5). Mod IV is within 0.02 to 0.04 and 13 to 20 % roll. Phugoid frequency in the airspeed mode is 33 % off at 88 KCAS (0.26 against 0.19 rad/s), where the simulator's figure is off its own trend of 0.24 and 0.25. Those limits were written for a table model, and some still print small misses. In-plane propeller force causes all but that one: it's real, the simulator may not have it, and it stays. At 80 KCAS, Mod III's stall speed, Dutch roll falls to 0.04 against 0.11. The table model gives 0.06 there, so that follows the CFD's loss of directional stability, not the built-up model.

5.5 High lift

(C) With the CFD's thrust, Mod IV at 58 KCAS and flaps 30 reaches $C_{L\max} = 4.54$ (4.40–4.58), the propellers adding 1.23 to 1.67. Figure 6 and table 3 show the three slipstreams tried. With no constant at all the ring wake halves the plain jet's error and brings moment and drag to their limits. Lift stays about 0.3 high, and one constant takes it inside the solvers' spread, as well as the three it replaced.

5.6 A motor lost

(S) The published failure, no pilot, 6000 ft, 95 KCAS [12], is described as takeoff flaps and gear down. Its traces can't be that. Its trim incidence of 8.4° and Mod II's 5.4° both sit on the cruise line of the Mod II paper's figures, and its airspeed falls at the rate of about 115 lbf lost, where takeoff flaps and gear need 180 a motor. The table model, flown clean, lands on every trace at once (fig. 7). This is an inference from the published numbers, not something stated. The test now flies clean, with its targets and limits as they were, and the takeoff-flap run still prints.

Table 3: Slipstream models at 58 KCAS, flaps 30.

	Constants	$C_{L\max}$	C_L error
Limit		4.40–4.58	0.078
Jet, as first fitted, discs alternating	3	4.52	0.078
Jet, no fit	0	5.55	0.690
Ring wake, no fit	0	4.99	0.327
Ring wake	1	4.54	0.072

Flown clean, the built-up model at first rolled too little, −60° at 3 s where the table model gave −81°. That was a real departure from the CFD, in two parts: sideslip's roll near stall, and one propeller's own roll. With both fitted it gives bank −79°, pitch −10.8°, heading −23.2°, peak roll rate −49 deg/s and sideslip 10.4°, sink 20.6 ft/s, against about −87°, −11.5°, −23°, −47, 9.1°, 24.5. The check that first flagged the sideslip as twice the published figure was comparing unlike configurations.

5.7 Climbs, control and failures

(S) With a motor out and feathered Mod III sinks 437 ft/min (450), and 352 (300) with a traction bus out. Minimum control speed is 81.7 KCAS (85), set by the rudder. Every row of the published failure list [10] is present and does what the row says. A contactor failure at 58 KCAS costs ΔC_L −0.30 and −0.34 (read as −0.30 and −0.35) **(C)**.

Five climbs miss. Mod III at full power climbs 515 ft/min (430), and at 2400 rpm 344 (270), while the motor-out and nominal DiTTo cases pass: gear drag can't serve all four. DiTTo with a motor out gives 69 and 435 ft/min (360 and 530). The published drop at 97 KIAS implies 108 lbf lost from a motor whose published tables give 173, and the table model drops by the same as the built-up one, so this isn't the built-up model's. The arithmetic is short: climb lost is thrust lost times true airspeed over weight, so about 170 lbf net at 170 ft/s on 3075 lb costs some 560 ft/min before any rudder and aileron drag. The model loses 643, NASA 360. On the real P2006T one engine out costs 81 % of the climb [18]; here it's 90 % at 97 KIAS and 59 % at 84, against NASA's 50 and 47. That comparison is loose, since a tip motor is about 30 % of the thrust and not half, and NASA's simulator may hold an effect its published tables don't show. DiTTo with a bus out gives −208 (−100).

(C) Ground effect is fitted, so flaps 30 no longer tests anything. Flaps 10's worst points stay at 0.063 and 0.123, and with the propellers on 0.152 and 0.296, speeds averaged.

5.8 Wing bending

(M) Fitted to the vibration test's first modes (2.25 and 6.55 Hz, 2.14 antisymmetric). Unfitted: clean antisymmetric 6.05 Hz (5.76), clean second bending 27.75 (26.36) and 25.70 (23.90). With the tip masses second bending misses, 18.50 (15.76) and 17.06 (14.82): the nacelle ahead of the wing's axis couples it with torsion, which a beam lacks. The model only flies the first mode. **(S)** Whole aircraft 2.39 Hz (2.31) for Mod III and

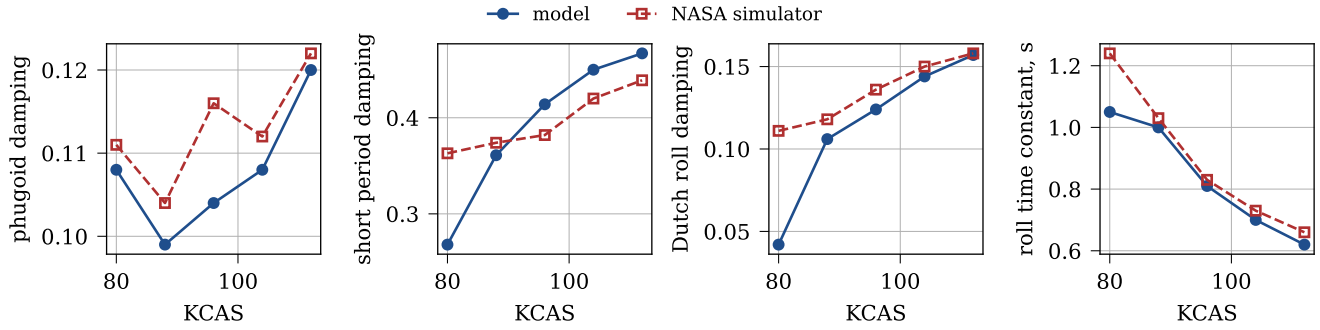


Figure 5: Mod III's modes against the simulator's **(S)**.

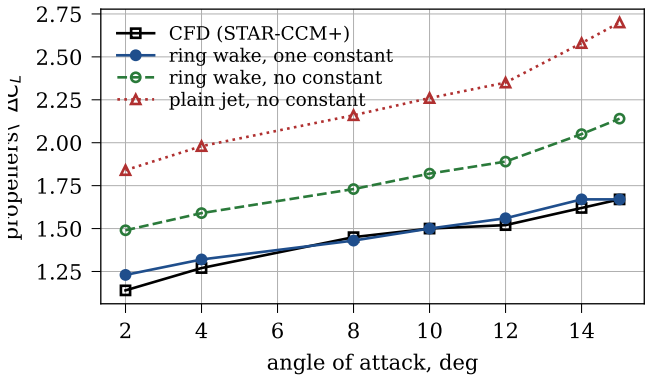


Figure 6: Lift the high-lift propellers add, with the CFD's thrust **(C)**.

2.37 (2.29) for Mod IV. In JSBSim with the air nearly gone the modes ring at the beam's frequencies within 1 %; in flight they're damped at every speed to V_{NE}. There's no torsion, so no flutter prediction. Bending off gives the rigid model's forces exactly.

5.9 Cockpit and FlightGear

(D) The avionics buses, sheds and propeller modes pass against the handbook. In FlightGear each overhead and lower switch, the wired breakers, light tests, alerts, manual pitch, the lever's feather stop, the high-lift logic, flaps, Autostart and Autostop were driven by the property each control sets. All 113 clickable controls have their objects and Nasal. Hands off in cruise it holds height within 3 ft in 60 s. Real mouse clicks were not automated.

5.10 Sound

Every sound plays on one scale, a full-scale sample at volume 1 being 112.5 dB re 20 μPa, in the cockpit and outside alike. The propellers, motors, the wind on the windscreen and the gear in the airflow are computed by PyWopWop [25], the last by Fink's method [24] as for the ASK 21's fixed wheels: at 90 kt the gear is 66.7 dB(A) at 30 m, about one cruise propeller's level, and 73.3 in the cockpit, some 16 dB under the wind. Mods III and IV were never heard, and no recording of the P2006T was used, so all of this is computed, not checked; the motors' levels and the wind's 91 dB(A) at 110 kt are the least certain. The gear's pump and locks, the flap motor, the tyres and the touchdown use stock samples at stated levels, with the tyres from tyre and road data and the rest estimates: none has a measurement behind it on this aircraft. The aural alert is set 15 dB over the cockpit's noise at 100 kt, after ISO 7731. A check reads each level back from the sound file.

6 Corrections found during validation

- A blade-element solver took the first root it met, and lightly loaded sections have false ones. Phugoid damping, which hangs on how thrust moves with speed, exposed it.
- The rudder scale sat ahead of a table ending at 26°, stopping it at 17°. Later the fin's table itself proved a quarter weak at full travel.
- A wing's divide by a speed near zero, and JSBSim's $b/2V$, threw the aircraft into NaN at rest with the rotors turning. Both are floored at 30 ft/s.
- Reading JSBSim's accelerations for the panels' coupling gave wrong frequencies and negative damping; it's solved in the system now. Raw roll rate fed to the ailerons pumps the 3.55 Hz mode up, so a test pilot's rate passes a 0.25 s lag.
- The force graph couldn't start in FlightGear until it became a system channel.
- FlightGear's default electrical system was running beside the model's and overwriting one voltage. It's off.
- The door handles called a module the package didn't load.
- The high-lift propellers first alternated along each wing, from a loose reading of the simulator paper. Turned as the geometry and the CFD have them, maximum lift rose by 0.06 with no constant changed, a contactor failure's yaw came inside its limit, which had been a miss, and the propellers' sound outside moved by up to 10 dB, because the side view hears the rising blades.
- The cruise blades in the visual model were pitched for the opposite rotation to their animation.
- The failure test ran the wrong configuration (section 5.6), and minimum control speed was read from an unsolved point.
- The gear retracted on the ground. The flight manual shows only a knob to pull first, and other sources differ on a squat switch. A guard was added as a simulator choice.

7 Assessment

41 checks miss. They fall into four kinds:

- **A source's figure in doubt.** The P2006T flight paper's short period and lift and moment slopes. That aircraft flies, so the reasons given can be weighed against other flight data.

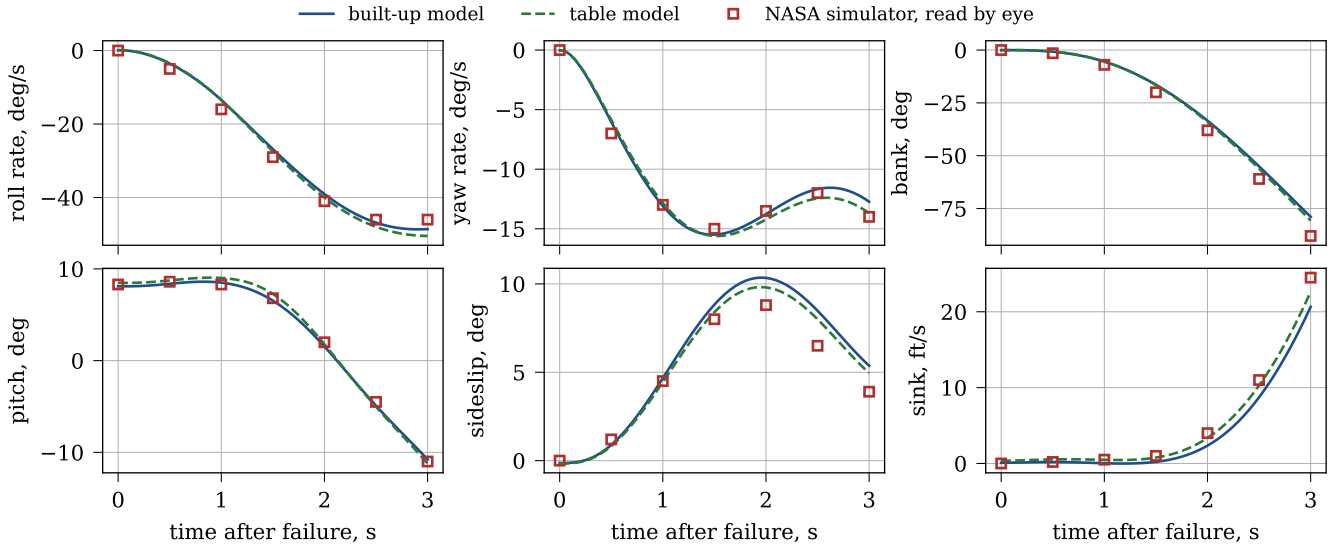


Figure 7: The left motor lost at 6000 ft, 95 KCAS, no pilot, flown clean **(S)**.

- **Sources that can't all be met.** Mod III's two climbs against its motor-out and DiTTo climbs; the DiTTo motor-out climbs against the published motor tables, a mismatch inside the published figures that no flight can settle; the Mod II simulator's Dutch roll and phugoid against the flight test.
- **Limits or tests of the author's own.** Parked bank −0.17° against 0.1, from the CG sitting left; P2006T landing roll 138 m (209), on an assumed full braking; several mode limits written for the table model.
- **Real or open.** Ground effect at flaps 10 and with the propellers on; a 12 % high lift slope on the P2006T; the aileron's pitching moment and power-on roll from sideslip; second bending with tip masses; $C_{Y\beta}$; the Mod II simulator's roll time constant.

The motors' and rotors' inertias, loads, the torque rate and the speed-loop gains are estimates. Not modelled: torsion and flutter, a roll or yaw rate's crossflow at the discs, dynamic inflow, the motors' temperatures. Mods III and IV never flew, so a pass against CFD or a simulator shows agreement with a prediction and no more, and a miss doesn't show which of the two is wrong. This work isn't connected with or endorsed by NASA, Tecnam or the aircraft's makers.

8 Conclusions

- A built-up model of the X-57 Mods III and IV, each of its fourteen propellers a disc with its own speed, its wing strips in their own slipstreams and its wing bending, runs in plain JSBSim at 120 Hz, generated from published data.
- Proved on the P2006T it passes the flight manual, and a second source shows the flight test's short period to be a method artefact.
- A ring-vortex slipstream with one constant does as well as a jet with three, and with none it halves the error of a plain jet.
- The published motor failure is reproduced in time history, once flown in the clean configuration its own traces show.
- Limits were set first and never moved. 199 checks pass and 41 miss, each with its cause or marked open.

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